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# Bioacoustic signal denoising: a review

Jie Xie<sup>1,2,3</sup> · Juan G. Colonna<sup>4</sup> · Jinglan Zhang<sup>5</sup>

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## Abstract

Animal biodiversity has been experiencing rapid decline due to various reasons such as habitat loss and degradation, invasive species, and environment pollution. Recent advances in acoustic sensors provide a novel way to monitor animals through investigating collected bioacoustic recordings. To accurately monitor animals, the precondition is the high performance of developed bioacoustic signal recognition model. However, since bioacoustic recordings are often obtained in an open environment, various sources of noise will affect the audio quality, which causes problems for automated analysis of animal sound recordings. Although various methods have been developed for addressing the noise in different bioacoustic recordings, to the best of our knowledge, there is still no paper that reviews and summarizes those methods. The main aim of this paper is to provide a systematic survey of the existing literature related to bioacoustic signal denoising. By investigating the existing denoising methods for bioacoustic recordings, current challenges, possible opportunities, and future research directions are discussed and concluded.

**Keywords** Bioacoustic signal denoising · Bioacoustic signal enhancement · Noise reduction

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✉ Jie Xie  
xiej8734@gmail.com

Juan G. Colonna  
juancolonna@icompu.ufam.edu.br

Jinglan Zhang  
jinglan.zhang@qut.edu.au

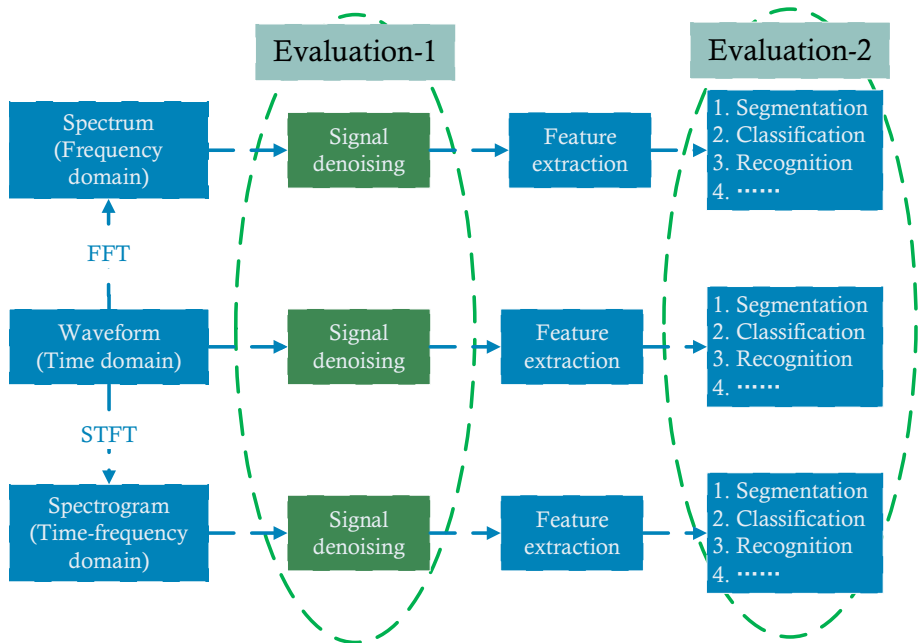
<sup>1</sup> Key Laboratory of Advanced Process Control for Light Industry (Ministry of Education), Jiangnan University, Wuxi 214122, People's Republic of China

<sup>2</sup> School of Internet of Things Engineering, Jiangnan University, Wuxi 214122, People's Republic of China

<sup>3</sup> Jiangsu Key Laboratory of Advanced Food Manufacturing Equipment & Technology, Jiangnan University, Wuxi 214122, People's Republic of China

<sup>4</sup> Instituto de Computação (IComp), Universidade Federal do Amazonas (UFAM), Av. General Rodrigo Octávio, 6200, CEP: 69080-900, Manaus, Amazonas, Brasil

<sup>5</sup> School of Computer Science, Queensland University of Technology, Brisbane, Australia



**Fig. 1** Bioacoustic signal denoising. Here, signal denoising is applied to audio data in three domains: time domain (waveform), frequency domain (spectrum) and time-frequency domain (spectrogram). The spectrum and spectrogram are generated by applying fast Fourier transform and short-time Fourier transform to waveform, respectively. For the evaluation of signal denoising method, the first method (*Evaluation-1*) is to directly measure the signal difference before and after denoising. The second method (*Evaluation-2*) is based on the performance of the whole system, where the effect of signal denoising is often not explicitly investigated in previous studies

## 1 Introduction

Animal biodiversity has been experiencing rapid decline due to various reasons such as habitat loss and degradation, invasive species, and environment pollution (Xie et al. 2018). Recent development of acoustic sensors provide a novel way to monitor animals by collecting bioacoustic recordings. Through investigating collected recordings, ecologists can study ethology of animal acoustic communication, evolution, ontogeny, and development of acoustic behavior, relationships between animal sounds and their environment, effects of man-made sounds on animals (Fletcher 2007; Deichmann et al. 2018). However, all those ecological research goals have a precondition that those bioacoustic recordings are well investigated such as the animal species can be accurately recognized. Taking animal sound classification for example, it often has three processing steps: signal preprocessing, feature extraction, and classification (Fig. 1). From Fig. 1, we can observe that the result of signal denoising has a direct affect on the result of feature extraction, and then segmentation, recognition or classification. For signal preprocessing, signal segmentation and denoising are two main procedures, where the use of bioacoustic signal denoising is highly dependent on the recording quality. If the bioacoustic recordings are obtained in the laboratory, there might be no need to use denoising methods because of the high signal-to-noise ratio (SNR) of those recordings. However, most bioacoustic recordings are obtained in an open environment, therefore, developing automatic methods

for reducing the noise in those bioacoustic recordings is in high demand. Various bioacoustic signal denoising methods have been developed and can be split into three main types based on the signal representation type: time domain, frequency domain, and time-frequency domain.

Since both bioacoustic signal denoising and speech enhancement aim to improve the SNR, many methods, which were designed for speech enhancement have been successfully adapted for bioacoustic signal denoising. However, there are obvious differences between recordings of bioacoustic signal and speech: (1) the signal source of bioacoustic recordings is dependent on animals of interest. For instance, when bird and cicada co-exist in the same recording, bird sounds are of interest, then cicada sounds will be regarded as the noise and vice versa; (2) the sources of noise for bioacoustic recordings by geophony (environmental sounds such as wind and rain), anthrophony (noise generated by humans, though sources such as traffic and machines), and biophony (sounds from animals that are not of interest) (Alonso et al. 2017; 3) the SNR of bioacoustic recordings is typically low, particularly if the microphone of acoustic sensor is not very close to acoustic sources; (4) discriminative information exists in various frequency ranges which depends on the studied animal, and thus there is no specific structure for bioacoustic signals like speech which has a grammar-like structure (Chandrakala and Jayalakshmi 2019). According to those differences, various methods have been developed aiming to increase the SNR of bioacoustic recordings, then further improve the performance of subsequent analysis such as species classification, population estimation, etc. The role of signal denoising played in a system is shown in Fig. 1.

Bioacoustic signal denoising often plays an important role in a whole animal sound recognition or classification system, but are among the least researched (Priyadarshani et al. 2018). To date, many bioacoustic signal denoising methods have been investigated in the context of different animals, thus a survey of bioacoustic signal denoising techniques is highly needed.

The primary contribution of this work is the collection and grouping of bioacoustic signal denoising methods, which can not only let experienced researchers know the hotspots in their area but also provide the newcomers the active research area to quickly follow. Another contribution is to summarize the methods used for different animal species and provide newcomers on quickly selecting an appropriate method to start the research.

In this survey, we primarily focus on bioacoustic signal denoising methods, which are often used as a preprocessing tool for improving the performance of subsequent analysis including signal segmentation, feature extraction, recognition or classification. Bioacoustic signal denoising is applied to three types of audio data representations: waveform, spectrum, and spectrogram, where spectrum and spectrogram are generated by applying fast Fourier transform (FFT) and short-time Fourier transform (STFT) to waveform. Six types of bioacoustic signal denoising algorithms are summarized.

The remainder of this survey is organized as follows: Section 2 outlines the methods for bioacoustic signal denoising. Section 3 describes the metrics used for evaluating bioacoustic denoising methods. Section 4 discusses the state-of-the-art denoising methods for different animal species. Finally, experimental results, discussion and conclusion are given in Sects. 5, 6, and 7, respectively.

## 2 Bioacoustic signal denoising

Various bioacoustic signal denoising methods have been developed and can be grouped into three domains: time domain, frequency domain, and time-frequency domain, including (1) Optimal FIR filter, (2) spectral subtraction, (3) minimum-mean square error short-time

spectral amplitude estimator (MMSE-STSA), (4) wavelet-based denoising, (5) image processing based noise reduction, (6) deep learning based noise reduction.

For bioacoustic signal, the call of a particular animal species often occupies a certain frequency range, which make it possible to eliminate the information that is not within the target frequency range (Brown et al. 2017). Take bird calls for examples, the frequency range is often between 1 kHz and 12 kHz (Härmä 2003; Pijanowski et al. 2011). Therefore, we can directly remove the frequency information that are not between 1 kHz and 12 kHz. In particular, three types of filtering methods are designed including low-, band-, and high-pass filtering, which are often used as the preprocessing steps in bioacoustic signal denoising.

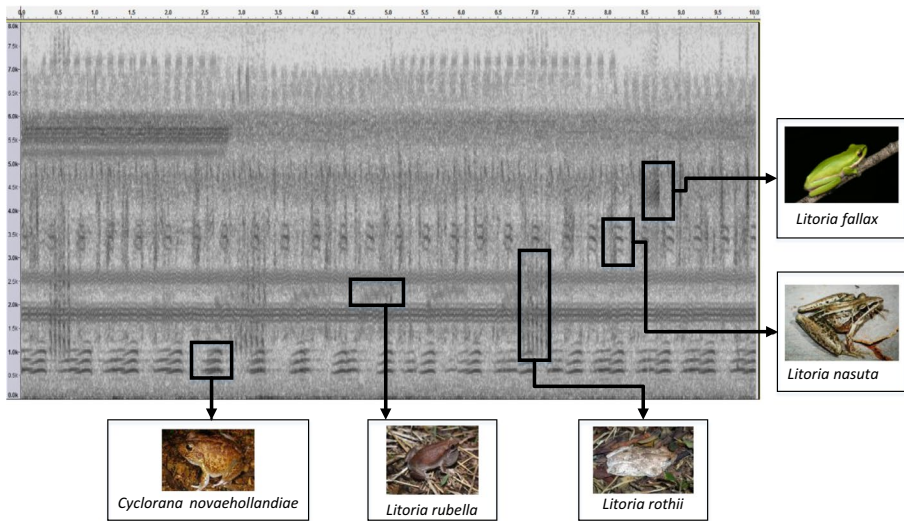
In audio applications, a high-pass filter passes signals with a frequency higher than a certain cutoff frequency and attenuates signals with frequencies lower than the cutoff frequency. A low-pass filter passes signals with a frequency lower than a selected cutoff frequency and attenuates signals with frequencies higher than the cutoff frequency. A band-pass filter passes frequencies within a certain range and rejects (attenuates) frequencies outside that range. A high-pass filter can be used in conjunction with a low-pass filter to produce a band-pass filter. Here, the aim of the filtering methods is to separate interested animal calls from the background noise.

A summary of those pass filtering methods is shown in Table 1. From the table, we can find that all those methods are highly dependent on the investigated animal species. When the frequency of studied bioacoustic signal is high, a high-pass filter will be used. When the frequency of studied bioacoustic signal is low, a low-pass filter will be used. However, if a bioacoustic recording has different species, that are of interest, whose frequency ranges are overlapped, this method cannot successfully denoise the bioacoustic signal. Fig. 2 shows an example of simultaneously vocalizing frog calls. We can find that the frequency of *Cyclorana novaehollandiae* and *Litoria rubella* are overlapped with *Litoria rubella*. Simply using pass filtering methods cannot successfully reduce the noise of collected recordings.

## 2.1 Optimal FIR filter

In this subsection, two types of FIR filters are described: Wiener filter and Singular Spectrum Analysis (SSA) filter. Wiener filtering often aims to optimize the minimum mean square error between the target signal and the predicted signal. Previous studies have investigated Wiener filtering for speech enhancement Li et al. (2011), where 1D and 2D Wiener filtering can be used separately or together for reducing the noise (Ding et al. 2009). For bioacoustic signals, 1D Wiener filtering is directly applied to waveform, while 2D Wiener filtering is applied to the spectrogram, which is generated by performing short-time Fourier transform on the waveform. In addition, 2D Wiener filtering has been used in previous studies (Hussein et al. 2012; Towsey et al. 2012; Pourhomayoun et al. 2013). Hussein et al. (2012) compared various noise reduction methods for producing the enhanced spectrogram, which were used for bioacoustic call classification including bats and birds. Here, Wiener filtering was employed for spectrogram enhancement.

Wiener filtering assumes that both signal and noise are stationary and the spectral information is available, which is not necessarily true for bioacoustic signal, especially when dealing with long duration bioacoustic recordings. In addition, when dealing with bioacoustic recordings with unknown noise, the noise level can be easily underestimated and the enhancement can be slightly milder (Kim et al. 2000; Hussein et al. 2012).



**Fig. 2** An example of collected recording with multiple simultaneously vocalizing frog species (Xie et al. 2018). Five frog species exist in this 10-second recording: *Cyclorana novaehollandiae*, *Litoria rubella*, *Litoria nasuta*, *Litoria rothii*, and *Litoria fallax*

Different from Wiener filter which minimizes the error using a reference signal, the Singular Spectrum Analysis (SSA) filter uses principal component analysis's information compression principle. This means that the SSA uses its own input signal to find the filter coefficients. The main difference between the SSA filter and the Wiener filter is the way in which the filter coefficients are obtained. Also, a difference in SSA is that it does not assume the stationary of signal. Colonna and Nakamura (2018) proposed a non-parametric bioacoustic signal filtering method based on SSA and several entropy criteria. Using the proposed method, an optimal finite impulse response filter bank was built for bioacoustic signals enhancement. An example of SSA-based filter for denoising anuran recordings is shown in Fig. 3.

## 2.2 Spectral subtraction

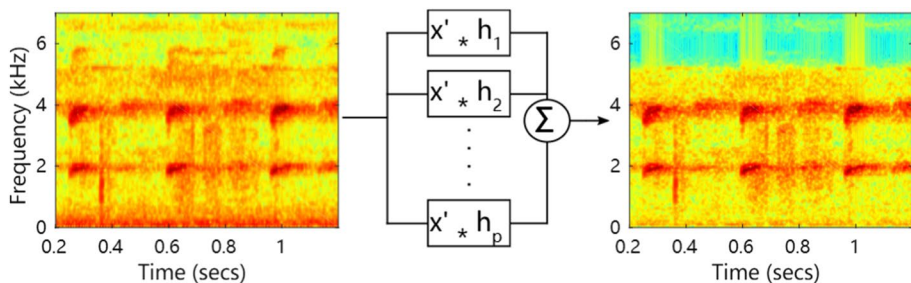
Spectral subtraction is one of the most common approach for audio noise reduction (Boll 1979). To be specific, a “noise profile” is first generated using a particular audio with only noise. Then, the noisy component of the signal is analyzed to be split into several spectral parts. Finally, the “noise profile” is subtracted from the entire audio file and only signal is theoretically remained. Here, the choice of suppression factor for “noise profile” needs to be carefully considered for balancing the noise and signal (McAulay and Malpass 1980).

Towsey et al. (2012) reduced noise of bioacoustic recordings by removing the modal noise intensity calculated for each frequency band. The method was a modified version of the same *adaptive level equalization* algorithm developed by Lamel et al. (1981). Patti and Williamson (2013) used spectral subtraction as a preprocessing step for improving bird species classification performance, but the effectiveness of noise reduction was not investigated. Bedoya et al. (2014) proposed to reduce the noise of anuran recordings by estimation and suppression of undesired component in the selected frequency band

**Table 1** A brief summary of low-, band- or high-pass filtering methods in the literature.

| Animal species      | Low-pass | Band-pass | High-pass | Combined method  | Reference                   |
|---------------------|----------|-----------|-----------|------------------|-----------------------------|
| Bird                |          |           | ✓         |                  | Baker and Logue (2003)      |
| Bird                |          |           | ✓         |                  | Neal et al. (2011)          |
| Bird                |          |           | ✓         |                  | Brown et al. (2017)         |
| Bird, whale         |          |           | ✓         |                  | Roger et al. (2018)         |
| Frog                |          |           | ✓         |                  | Chen et al. (2012)          |
| Frog                |          |           | ✓         |                  | Huang et al. (2014)         |
| Frog                |          |           | ✓         |                  | Xie et al. (2016a)          |
| Bat                 |          |           | ✓         |                  | Henríquez et al. (2014)     |
| Delphinids          |          |           | ✓         |                  | Simões Amorim et al. (2019) |
| Whale               |          | ✓         |           |                  | Zaugg et al. (2010)         |
| Whale               |          | ✓         |           |                  | Bermant et al. (2019)       |
| Bird, frog, cricket |          | ✓         |           |                  | Brandes (2008)              |
| Bird                |          | ✓         |           |                  | Knight et al. (2019)        |
| Bird                |          | ✓         |           |                  | Barmatz et al. (2019)       |
| Bird                |          | ✓         |           |                  | Baker and Logue (2007)      |
| Bird                | ✓        |           |           | Noise gating     | Bardeli et al. (2010)       |
| Bird                |          |           | ✓         | MMSE-STSA        | Brown et al. (2017)         |
| Bird                |          |           | ✓         | MMSE-STSA        | Brown et al. (2019)         |
| Bat                 |          |           | ✓         | Wiener filtering | Henríquez et al. (2014)     |
| Bird                | ✓*       | ✓*        |           | WPD              | Priyadarshani et al. (2016) |

Here, WPD denotes wavelet packet decomposition, MMSE-STSA denotes minimum-mean square error short-time spectral amplitude estimator, \* denotes that one filtering method is selected for being combined with WPD

**Fig. 3** Original and filtered spectrogram with an eigenfilter FIR bank built from the first four SSA principal components (Colonna and Nakamura 2018)

spectrum of the signal. Two main steps were included in the noise reduction: threshold estimation and noise removal. Xie et al. (2016b) used spectral subtraction for improving the syllable segmentation result for the environmental recording, which can sequentially improve the classification performance of frog calls.

For all those spectral subtraction related methods, a noise profile needs to be first estimated and the performance of spectral subtraction is highly dependent on the noise profile. However, the background noise of diverse environmental recordings are not consistent, it is difficult to use one noise profile to cover all types of background noise. In addition, the signal is assumed to be additive and uncorrelated with the signal (Pandey et al. 2004). Moreover, spectral subtraction often produces sounds with musical artifacts which are often more objectionable than the original noise. Therefore, the spectral subtraction method cannot deal with long duration environmental recordings with rapidly changed background noise. Cai et al. (2007) proposed a method where the noise profile was estimated from every frame. However, this method only worked when the analyzed signal was dominant in the recording with a fixed duration.

### 2.3 Minimum-mean square error short-time spectral amplitude (MMSE-STSA) estimator

MMSE-STSA estimator is developed by Ephraim and Malah (Ephraim and Malah 1984) for speech enhancement. A near-optimal short-time spectral amplitude estimator is used to significantly improve noise reduction by reducing the artifacts. To further improve the performance of MMSE-STSA, Ephraim and Malah (1985) used log spectral.

Similar to other speech enhancement methods, MMSE-STSA has been successfully applied for denoising bioacoustic recordings. Alonso et al. (2017) used MMSE-STSA for helping the semi-automated segmentation of anuran calls. Then, the segmented anuran calls were classified using machine learning algorithms. Brown et al. (2017) proposed using the MMSE-STSA method with band-pass filtering for automated denoising of environmental recording for bird sound identification. The MMSE-STSA method was compared with a wavelet transform based denoising method proposed by Priyadarshani et al. (2016) in terms of *Composite Evaluation Measures* (Hu and Loizou 2006) and efficiency.

MMSE needs to perform the noise estimation, which is usually updated from noise-only frames. Very often noise-only frames are determined by voice (signal) activity detectors (VADs). However it is unreasonable to assume that the first few frames of the environmental recordings are noise only without animal calls. Sometimes there is almost no gap between animal calls; therefore there are not enough noise-only frames to update noise estimation. Moreover, VADs do not work well in low SNR signals (Cai et al. 2007).

### 2.4 Wavelet-based denoising

Wavelet-based denoising method is another collection that has been widely used in bioacoustic signal denoising including manatee (Gur and Niezrecki 2007, 2011), owl (Gómez et al. 2018), anuran (Huang et al. 2014), rhesus monkey (Ren et al. 2008), and humpback whale (Ren et al. 2008), and other birds (Sun et al. 2013; Ren et al. 2008; Priyadarshani et al. 2016).

Ren et al. (2008) proposed a method based on a perceptually scaled wavelet packet decomposition using a species-specific Greenwood scale function. Spectral estimation techniques, similar to those used for human speech enhancement, were used for estimation of clean signal wavelet coefficients under an additive noise model. Experimental results, measured by both SNR and segmental SNR of the enhanced wave forms, indicated that their proposed method outperforms other approaches for a wide range of noise conditions. In addition, several animal species were used for evaluating including the ortolan bunting

(*Emberiza hortulana*), rhesus monkey (*Macaca mulatta*), and humpback whale (*Megaptera anovaeanglia*). Gur and Niezrecki (2007, (2011) developed two wavelet based methods for enhancing manatee vocalizations. The first method explored both the autocorrelation function of manatee vocalizations and effectiveness of the nonlinear wavelet transform based denoising algorithms. For the second method, they developed an adaptive denoising procedure, implemented on the wavelet packet transform coefficients obtained from the noisy vocalization signals. Sun et al. (2013) explored two wavelet transform based method for bird sound denoising: soft threshold and hard threshold using Matlab build-in function, where hard threshold was found to be better in denoising signals of 13 bird species. Huang et al. (2014) used the wavelet threshold function as the denoise filter kernel function for removing noise of anuran recordings in 1D signal.

Recently, Priyadarshani et al. (2016) proposed a method of denoising using a combination of the wavelet packet decomposition and band-pass or low-pass filtering. For wavelet packet decomposition, the selection of Mother wavelet, decomposition level, and threshold were investigated for optimizing the denoising algorithm. Gómez et al. (2018) proposed a Last Approximation standard deviation algorithm for the processing of bioacoustic signals (owl) based on wavelet analysis. The proposed method was found to be better than a standard denoising method based on the Matlab Wavelet Toolbox.

Those methods based on wavelet packet decomposition or wavelet transform are effective in removing background noise in the spectrogram. However, the noise generated during the Fourier transform is difficult to be suppressed, since the former noise is usually random Gaussian distribution while the latter may be modeled by Rayleigh distribution (Hussein et al. 2012).

## 2.5 Image processing based noise reduction

After transforming 1D bioacoustic signal (waveform) into its 2D representation (spectrogram), image analysis techniques which treat the spectrogram as an image provide a range of methods for reducing noise in bioacoustic recordings.

Hussein et al. (2012) used the edge detection techniques to produce an enhanced spectrogram. The operation was based on spectrogram with the limited dynamic power range. One edge detection algorithm was used to detect sound patterns by reconstructing the original power values of the pattern edges and their interior, but the power values of the patterns surrounding were eliminated which were regarded as the attached noises. Zeppelzauer et al. (2013) developed a method for enhancing the spectrogram using structure tensor, which was subsequently used for acoustic detection of elephant presence. In the work of Lin et al. (2013), the pre-whitened spectrogram was convolved using a 3×3 Gaussian kernel to produce a smoothed spectrogram, which aimed to conserve tonal sound edges and reduce isolated noise level concurrently. It is worth noting that the pre-whitened spectrogram was obtained by subtracting the power spectrum to an exponentially weighted running mean for each frequency bin (Mellinger 2004; Baumgartner and Mussoline 2011).

Recently, Esfahanian et al. (2017) proposed a method to pre-process whale vocalization spectrograms for noise reduction and tone removal, where non-upcalls were eliminated by an energy detection algorithm. Koluguri et al. (2017) proposed a spectrogram enhancement technique using a multiple window Savitzky–Golay (MWSG) filter for improving bird sound detection. With the proposed method, the spectrogram estimate using MWSG filter was unbiased and had lower variance compared with its single window counterpart. Heim et al. (2019) applied a Savitzky–Golay filter with a filter window length of 13 and a

polynomial order of three for reducing noise in each echolocation call file. Compared to the spectrogram-based denoising algorithm, the use of variant maps led to the performance improvement of some bat species, but worse performance for other species. Therefore, a combination of variant maps and spectrogram can significantly improve the automated identification of species based on acoustic information.

## 2.6 Deep learning based noise reduction

Deep learning has shown a great success in various pattern recognition tasks including image denoising (Zhang et al. 2017; Yang et al. 2018; Lefkimmiatis 2018) and speech enhancement (Lu et al. 2013; Fu et al. 2016; Rethage et al. 2018; Dionelis and Brookes 2019). Lots of works have applied deep learning for bioacoustic signal detection and classification, which achieves the state-of-the-art performance (Bergler et al. 2019; Oikarinen et al. 2019; Stowell et al. 2019). Compared with traditional denoising algorithms that depend on signal models and their corresponding prior assumptions, removing noise with a deep learning method need to be trained using a large training set. After the completion of training, the deep learning method achieves adaptive denoising without requirements of (1) Accurate modelings of the signal and noise, and (2) Optimal parameters tuning (Yu et al. 2019). Therefore, using deep learning for bioacoustic signal denoising is worth being investigated, however, it is a challenge to obtain lots of tagged bioacoustic data, especially for different animal species under various background noise.

To the best of our knowledge, there is still no work that directly uses deep learning for bioacoustic signal denoising, which might be explained, to some extent, by the following work. Kong et al. (2017) indicated that noise reduction methods such as optimally-modified log-spectral amplitude, median filter and spectral subtraction did not help the detection performance of weakly labeled bird audio, where a joint detection and classification, convolutional neural network (CNN) model with VGG was proposed as a classifier, and CNN as a detector.

## 2.7 Summary

To better serve as a guild for newcomers to bioacoustic signal denoising, we present four selected denoising algorithms using a concise mathematical definition. Let  $y(t)$ ,  $x(t)$ , and  $d(t)$  be the original noisy signal, the clean signal and the noise, respectively, where the relation is

$$y(t) = x(t) + d(t) \quad (1)$$

The frequency domain formulation of the **Wiener filter** is given as

$$H(w) = \frac{S_{xx}(w)}{S_{xx}(w) + S_{dd}(w)} \quad (2)$$

where  $H(w)$  is the desired filter response and  $S_{xx}(w)$  and  $S_{dd}(w)$  are power spectral densities of the desired clean signal and noise. It is worthwhile noting that  $S_{xx}(w)$  is not directly available for a real bioacoustic recording.

For **spectral subtraction**, the direct subtraction with a mathematically optimal estimate for the signal spectrum in a MMSE sense is replaced by direct subtraction. The noise spectrum is estimated from the Fourier transform magnitude of a silence region

in the wave form, so that for each frame of the signal, an estimate for the clean signal in the frequency domain can be given directly as

$$\hat{X}(w) = [|Y(w)| - \hat{D}(w)]e^{j\theta_y(w)} \quad (3)$$

where  $\theta_y(w)$  is the phase component of the noisy signal, where the assumption is that the Gaussian additive noise does not change the phase of the original signal, it only changes the amplitude of the frequency spectrum, so the noise phase is considered linear. The signal phase is very important, which remains unchanged and does not distort the measured signal.

The **MMSE-STSA filter** is summarized using the equation for the minimum mean squared estimate of spectral amplitude of the clean signal

$$\hat{X} = G_{MMSE}(w)Y_w \quad (4)$$

where  $G_{MMSE}(w)$  is the spectral gain factor, which is denoted as

$$G_{MMSE}(w) = \frac{\sqrt{\pi v_w}}{2} e^{-\frac{v_w}{2}} \left[ (1 + v_w) I_0\left(\frac{v_w}{2}\right) + v_w I_1\left(\frac{v_w}{2}\right) \right] \quad (5)$$

where  $I_0(\cdot)$  and  $I_1(\cdot)$  are the modified Bessel functions of the zero-th and first order, respectively, and  $v_w$  is defined as

$$v_w = \frac{\xi_w}{1 + \xi_w} \gamma_w \quad (6)$$

where  $\xi_w$  and  $\gamma_w$  are estimated *a priori* and *a posteriori* SNRs for each spectral components.

**Wavelet-based denoising** aims to obtain the enhanced bioacoustic signals in wavelet domain, which is defined as

$$Y_{l,n}(k) = X_{l,n}(k) + D_{l,n}(k) \quad (7)$$

where  $Y_{l,n}(k) = \langle y, \pi_{l,n,k} \rangle$ ,  $X_{l,n}(k) = \langle x, \pi_{l,n,k} \rangle$ ,  $D_{l,n}(k) = \langle d, \pi_{l,n,k} \rangle$ ,  $k$  is the index of the coefficients in each subband,  $l$  is the decomposition level,  $n$  is the node index,  $\pi_{l,n,k}$  is the scaled and shifted mother wavelet. The notation  $\langle x, \pi \rangle$  represents the WT of signal  $x$  by using as the mother wavelet. Then, the estimated power of the enhanced bioacoustic signal in wavelet domain is given by

$$|\hat{X}_{l,n}(k)|^2 = \begin{cases} |Y_{l,n}(k)|^2 - |\hat{D}_{l,n}(k)|^2, & |Y_{l,n}(k)|^2 > |\hat{D}_{l,n}(k)|^2 \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

For image processing and deep learning based noise reduction, there is no uniform equations for description. Therefore, we do not provide the equations for describing those two types of denoising algorithms. Comparing those four selected denoising algorithms, wavelet-based denoising performs the noise reduction in wavelet domain, while others in spectral domain. Compared to spectral subtraction using the direct subtraction, Wiener filtering is conceptually similar but with a mathematically optimal estimate for the signal spectrum in a MMSE sense (Lim and Oppenheim 1978). Compared to Wiener filtering which is an optimal linear estimator of the clean signal spectrum in a MMSE sense, Ephraim and Malah extended this idea by deriving an optimal nonlinear estimator of the clean spectral amplitude (Ephraim and Malah 1984).

All six types of noise reduction methods have been widely explored in studying bioacoustic recordings. For each type of noise reduction method, we have reviewed the representative work and described their advantages and disadvantages. In addition to those aforementioned methods, Hu et al. (2005) designed a simple yet effective algorithm for canetoad monitoring. First, the recording was divided into a number of time slices which was one millisecond in length. If the amplitude level of the whole period was under a threshold (for example, from  $-20$  to  $+20$ ), they regarded it as a silent/noise only period which can thus be removed. Kandia et al. (2008) proposed the use of Teager-Kaiser energy operator for the enhancement of recordings made by hydrophones, where the final goal was to reduce the uncertainty in localizing sperm whales using passive acoustics. Xie et al. (2019) first introduced the average metropolis weight local average consensus algorithm to increase the consensus convergence rate, then a continuous spectrum update algorithm was proposed to estimate the noise power spectral density to improve the noise reduction performance. Lastly, a Minimum Variance Distortionless Response beamformer was introduced to enhance the bird audio. Recently, Lostanlen et al. (2019) proposed to perform unsupervised detection of bioacoustic events by pooling the magnitudes of spectrogram frames after per-channel energy normalization (PCEN). Here, PCEN was used for enhancing animal vocalizations.

In many cases, simply using one denoising method cannot successfully enhance the target signal. Therefore, many papers combined different noise reduction methods for improving the SNR of bioacoustic recordings. The advantages of each combined method can be utilized for improving the final denoising performance. Some combined methods are list in Table 1, where filtering methods are combined with other algorithms. Brown et al. (2017, 2019) proposed to combine MMSE-STSA method with high-pass filtering for denoising environmental recordings. Here, a 1 kHz high-pass filter was first applied, then MMSE-STSA was used to further improve the SNR of environmental recordings. Bardeli et al. (2010) combined low-pass filtering and noise gating method to improve the detection of *Eurasian bittern*. Towsey et al. (2012) combined a 2D smoothing Wiener filter with spectral subtraction for denoising. Pourhomayoun et al. (2013) used a 2D Wiener filter to denoise and smooth the spectrogram, where a  $5 \times 5$  window around each pixel was used to estimate the local variance. Henríquez et al. (2014) combined high-pass filtering with a Wiener filter to improve the segmentation and subsequent identification of bat calls. Priyadarshani et al. (2016) proposed a method of bird sound denoising, where band-pass or low-pass filtering was combined with the wavelet packet decomposition. It is worth noting that low-pass filtering is less frequently used, since most interested animal species in previous studies have a relatively high frequency. In addition to the combination of filtering methods, Lin et al. (2017) first pre-whitened the spectrogram by subtracting the power spectrum to an exponentially weighted running mean for each frequency bin. Then, a Gaussian kernel was used to produce a smoothed spectrogram.

### 3 Evaluation metrics

Although bioacoustic signal denoising has been widely used in bioacoustic signal analysis, very often they are used as a tool for helping other subsequent tasks within a whole system, such as syllable segmentation, feature extraction, and classification. The general effectiveness of those denoising methods are not analyzed in detail for environmental recordings. There are

mainly two ways to evaluate denoising methods, one involves human beings and one does not. It is possible to combine both type of evaluation metrics.

For the direct evaluation of bioacoustic signal denoising methods, some evaluation metrics have been proposed. Signal-to-noise ratios (SNR) and segmental SNR are two most widely used metrics. SNR is defined as the ratio of signal power to the noise power, and is often expressed in decibels. SSNR is computed by calculating the SNR on a frame-by-frame basis over the signal and averaging these values. This permits the measure to assign equal weights to the loud and soft portions of the signal, which has been shown to have a higher correlation with perceived quality in human speech evaluation (Deller et al. 2000). The formulas for SNR, SSNR, and SnNR are defined as follows:

$$\text{SNR} = 10 \log_{10} \frac{\sum_t \hat{y}^2(t)}{\sum_n [y(t) - \hat{y}(t)]^2} \quad (9)$$

$$\text{SSNR} = \frac{1}{M} \sum_{j=0}^{M-1} 10 \log_{10} \left( \sum_{t=Tj+1}^{T(j+1)} \frac{y^2(t)}{[y(t) - \hat{y}(t)]^2} \right) \quad (10)$$

$$\text{SnNR} = 10 \log_{10} \frac{\sum_t y^2(n)}{\sum_n [y(t) - \hat{y}(t)]^2} \quad (11)$$

where  $M$  is the number of frames, each of length  $T$  samples and  $y(t)$  and  $\hat{y}(t)$  are the original and enhanced signals.

Furthermore, two evaluation measures, success ratio and Peak Signal to Noise Ratio (PSNR) are used in Priyadarshani et al. (2016), and the definitions are given as follows:

$$\text{Success Ratio} = \log_{10} \left( \frac{\text{var}(d(t))}{\text{var}(y(t) - \hat{y}(t))} \right) \quad (12)$$

where  $d(t)$  is the original noise,  $y(t) - \hat{y}(t)$  denotes the reduced noise, and

$$\text{PSNR} = 20 \log_{10} \left( \frac{\text{MAX}_{\text{sig}}(y(t))}{\sqrt{\text{MSE}(y(t))}} \right) \quad (13)$$

where  $\text{MAX}_{\text{sig}}$  is the maximum value of the reference signal and MSE is the mean-squared error. In this calculation, the noisy recording is used as the reference and its enhanced recording as the test. PSNR will be relatively lower if the recording is less cleaned and higher if the recording is well cleaned.

In addition, composite measures are used for evaluating the signal denoising method involving subjective tests to assess the voice quality by human beings, where the measures consist of Signal Quality (SIG), Background Intrusiveness (BAK), and Overall Quality (OVL). The definition of those three metrics are

$$C_{\text{sig}} = 3.093 - 1.029 \cdot \text{LLR} + 0.603 \cdot \text{PESQ} - 0.009 \cdot \text{WSS} \quad (14)$$

$$C_{\text{bak}} = 1.634 + 0.478 \cdot \text{PESQ} - 0.007 \cdot \text{WSS} + 0.063 \cdot \text{SSNR} \quad (15)$$

$$C_{\text{ovl}} = 1.594 + 0.805 \cdot \text{PESQ} - 0.512 \cdot \text{LLR} - 0.007 \cdot \text{WSS} \quad (16)$$

where Weighted-Slope Spectral Distance (WSS) (Klatt 1982), Perceptual Evaluation of Speech Quality (PESQ) (Rix et al. 2001), Log Likelihood Ratio (LLR), and Itakura-Saito (IS) distance (Quackenbush 1995) are evaluated for filters and filter configurations.

Recently, since large volumes of bioacoustic data have been collected, another evaluation metric, efficiency has become an important evaluation metric for building real-time processing systems. It is worth noting that obtaining the truly clean recordings for evaluating the denoising methods is often difficult, especially studying environmental recordings. Previous studies often used an aggressive noise reduction approach to obtain the clean signal (Brown et al. 2017). However, it is difficult to assess the obtained clean signal.

## 4 Summary of bioacoustic signal denoising methods

Although many papers have used signal denoising methods for studying bioacoustic signals, the effects of those denoising algorithms have not been fully investigated. For instance, the SNRs before and after applying the denoising algorithm are not presented, but the bioacoustic detection or classification performance are given for evaluating the whole framework. Very few studies focused on the denoising algorithms, where they compared different denoising algorithms for finding the best one for specific applications (Gur and Niezrecki 2011; Hussein et al. 2012; Priyadarshani et al. 2016; Brown et al. 2017; Gómez et al. 2018). Even for those studies, different animal species were used as the research target, which makes it impossible for a direct comparison among those methods. A summary of those studies that focus on bioacoustic signals denoising methods are shown in Table 2. Here, AUDWT denotes autocorrelation based undecimated discrete wavelet transform; VISUSHRINK denotes a scheme that performs thresholding on the wavelet coefficients of the noisy signal; WPT denotes wavelet packet transform, WP-TNN denotes the wavelet packet thresholding neural network; FALE denotes feedback adaptive line enhancer filter; Lastd denotes a Last Approximation standard deviation algorithm. The equations for the evaluation can be found in Sect. 3.

From the table, we can find that the work among different authors are difficult to be compared. Different papers often use different animal species as their research targets. Then, according to the characteristic of those animal species, they design the specific denoising algorithms. It is difficult to argue that which algorithm is the best denoising algorithm, although the recent use of wavelet-based methods show its superior in denoising bioacoustic signals. However, wavelet-based methods are found to be less efficient in Brown et al. (2017). In their work, the execution time of MMSE-STSA is considerably shorter than wavelet-based methods.

Various bioacoustic signal denoising methods have been developed for different animal species. However, for a specific animal species, there is no information on which signal denoising method has been used. A summary of denoising methods for a specific animal species can be a good start for further investigation. Table 3 shows the representative methods for each specific animal species. It is found that wavelet-based denoising methods are the most widely used. In addition, bird species is the most interested animal species for investigating bioacoustic denoising algorithm.

**Table 2** A brief summary of studies that focus on the bioacoustic signal denoising methods, where *Evaluation-1* in Fig. 1 is used

| Animal species                                 | Propose method                      | Compared methods                                                                                 | Evaluation                | Reference                   |
|------------------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------|-----------------------------|
| Manatee                                        | AUDWT                               | Bandpass filter, VISUSHRINK(Donoho and Johnstone 1994), FIR-ALE and FALE (Yan et al. 2005, 2006) | SNR                       | Gur and Niezrecki (2007)    |
| Ortolan bunting, Rhesus monkey, Humpback whale | Perceptual WPT                      | Bandpass filter, Spectral subtraction, Wiener filter, Ephraim Malah, Uniform WPT                 | SNR, segmental SNR        | Ren et al. (2008)           |
| Manatee                                        | WP-TNN                              | FALE (Yan et al. 2006)                                                                           | SNR                       | Gur and Niezrecki (2011)    |
| Bird                                           | WPD and band-pass or lowpass filter | Different parameter settings                                                                     | SnNR, Success ratio, PSNR | Priyadarshani et al. (2016) |
| Bird                                           | MMSE-STSA                           | WPD (Priyadarshani et al. 2016)                                                                  | SIG, BAK, OVL             | Brown et al. (2017)         |
| Owl                                            | LAstd                               | Matlab Toolbox build-in function <i>wden</i>                                                     | SNR                       | Gómez et al. (2018)         |

**Table 3** Representative denoising algorithms for different animal species

| Target animal species | Type of denoising method | Reference                   |
|-----------------------|--------------------------|-----------------------------|
| Bird                  | MMSE-STSA                | Brown et al. (2017)         |
|                       | Wavelet-based            | Brown et al. (2017)         |
|                       | Wavelet-based            | Priyadarshani et al. (2016) |
|                       | Image-based              | Hussein et al. (2012)       |
|                       | Wavelet-based            | Ren et al. (2008)           |
| Anuran/Frog           | Spectral subtraction     | Bedoya et al. (2014)        |
| Manatee               | Wavelet-based            | Gur and Niezrecki (2007)    |
| Owl                   | LAstD                    | Gómez et al. (2018)         |
| Humpback whale        | Wavelet-based            | Ren et al. (2008)           |
| Rhesus monkey         | Wavelet-based            | Ren et al. (2008)           |
| Bat                   | Wiener filtering         | Hussein et al. (2012)       |
|                       | Image-based              | Henríquez et al. (2014)     |

## 5 Experiment results

Table 4 shows the denoising results of four bird species recordings, which were obtained from the Ecosounds website. The denoising algorithms used consist of Wiener filtering (Zavarehei 2020d), spectral subtraction (Zavarehei 2020a, b), MMSE-STSA (Zavarehei 2020c), and wavelet-based denoising (Islam et al. 2015). The codes of all algorithms can be obtained online, which are summarized in the Github<sup>1</sup> by the authors. From the table, we can find that all denoising algorithms can successfully improve the averaged SnNR. However, the performance of different bird recordings and evaluation metrics is very different. It is worthwhile noting that the parameter settings of all selected algorithms were set experimentally without fine tuning, since the goal of this paper is not to propose a novel bioacoustic signal denoising algorithm. Here, the experimental results and summarized codes are used to provide newcomers of bioacoustic signal denoising field a good start.

## 6 Discussions

In this section, the research gap of current bioacoustic signal denoising algorithms is identified to give a direction for future work. Bioacoustic signal denoising plays an important role in bioacoustic signal analysis, especially when analyzing environmental recordings collected by sensor networks, because the SNR of environmental recordings can vary from 30 dB to −10 dB mainly due to different weather conditions (Cai et al. 2007), the distances of birds from the sensors, and surrounding environments.

In most studies related bioacoustic signal analysis, signal denoising is used aiming to improve the performance of the subsequent analysis. However, the effect of using denoising algorithms is not fully investigated. For researchers, it is difficult to understand whether the performance of the whole framework will be improved when the denoising algorithm is used.

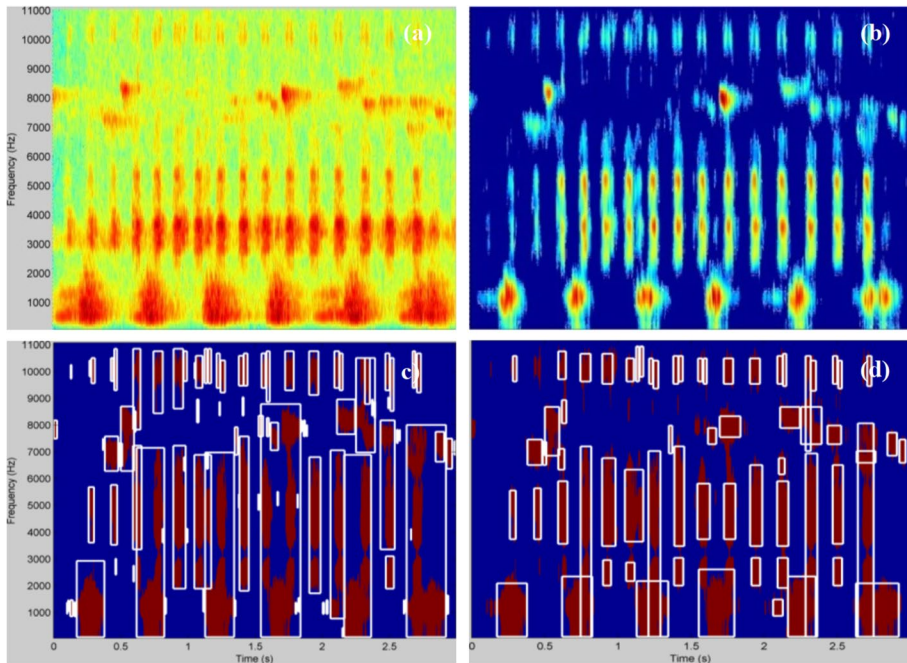
<sup>1</sup> <https://github.com/Jerry-Jie-Xie/BioacousticSignalDenoising>.

**Table 4** Experimental results of applying five denoising algorithms to recordings of four bird species

| Species           | Original                 | Wiener filter-1D     |         |           | Spectral subtraction (1) |         |                         | Spectral subtraction (2) |         |       |
|-------------------|--------------------------|----------------------|---------|-----------|--------------------------|---------|-------------------------|--------------------------|---------|-------|
|                   | Reference                | Zavarehei (2020d)    |         |           | Zavarehei (2020b)        |         |                         | Zavarehei (2020a)        |         |       |
|                   | SnNR                     | SnNR                 | S ratio | PSNR      | SnNR                     | S ratio | PSNR                    | SnNR                     | S ratio | PSNR  |
| Bush stone-curlew | 6.22                     | 2.54                 | 3.28    | 36.75     | 6.22                     | 2.78    | 37.02                   | 4.84                     | 0.55    | 40.98 |
| Grey fantail      | 2.01                     | 30.36                | 3.27    | 48.67     | 3.08                     | 0.56    | 51.41                   | 3.46                     | 0.48    | 51.54 |
| Southern boobook  | 6.94                     | 3.53                 | 3.20    | 40.24     | 6.94                     | 2.78    | 40.50                   | 5.47                     | 0.46    | 44.39 |
| Tawny grassbird   | 11.09                    | 23.15                | 2.68    | 60.47     | 6.02                     | 0.61    | 61.22                   | 9.88                     | 0.33    | 64.88 |
| Mean value        | 6.56                     | 14.90                | 3.10    | 46.53     | 5.56                     | 1.68    | 47.54                   | 5.91                     | 0.45    | 50.45 |
| Species           | Spectral subtraction (3) |                      |         | MMSE-STSA |                          |         | Wavelet-based denoising |                          |         |       |
|                   | Reference                | Towsey et al. (2012) |         |           | Zavarehei (2020c)        |         |                         | Islam et al. (2015)      |         |       |
|                   | SnNR                     | SnNR                 | S ratio | PSNR      | SnNR                     | S ratio | PSNR                    | SnNR                     | S ratio | PSNR  |
| Bush stone-curlew | 6.22                     | 8.75                 | 0.72    | 39.68     | 7.21                     | 1.26    | 38.42                   | 6.22                     | 2.98    | 37.14 |
| Grey fantail      | 2.01                     | 3.60                 | 0.79    | 47.46     | 10.72                    | 1.23    | 51.37                   | 3.96                     | 2.98    | 45.70 |
| Southern boobook  | 6.94                     | 11.02                | 1.14    | 42.61     | 4.27                     | 1.30    | 41.50                   | 20.11                    | 2.98    | 40.83 |
| Tawny grassbird   | 11.09                    | 11.03                | 0.61    | 62.78     | 11.87                    | 1.12    | 61.90                   | 19.46                    | 2.98    | 60.35 |
| Mean value        | 6.56                     | 8.60                 | 0.82    | 48.13     | 8.52                     | 1.23    | 48.30                   | 12.44                    | 2.98    | 46.01 |

Here, S ratio denotes success ratio

Low-, band-, and high-pass filtering are most widely used methods due to its simplicity. However, those methods cannot deal with recordings with overlapping calls and are highly dependent on the studied animal species. Therefore, some previous studies combined them with other signal denoising methods for achieving a better result (Bardeli et al. 2010; Brown et al. 2017, 2019; Henríquez et al. 2014; Priyadarshani et al. 2016). However, recordings with multiple simultaneously vocalizing animal species cannot be denoised by pass filtering, but have been investigated in other works. For instance, by learning from such weakly labeled data, a deep non-negative matrix factorization (NMF) model (Le Roux et al. 2015) can accurately recognize the spectral characteristics for bird and insect calls recorded in a forest. Besides NMF, source separation (SS) has been considered as a pre-processing step to detect cetacean vocalizations, fish choruses and shipping noise from estuarine and deep-sea soundscapes (Lin et al. 2017, 2019; Lin and Tsao 2019). Xie et al. (2015) used the image processing based method to separate different sound sources to extract features for frog call classification. Here, the image processing based method was used for separating different frog species. Fig. 4 shows an example of applying Wiener filtering, spectral subtraction (Towsey et al. 2012), and acoustic event detection (AED) for separating individual acoustic events. In addition, Ruiz-Muñoz et al. (2018) applied dictionary learning for spectrogram denoising in the presence of rain noise artifacts, where a dictionary from a clean set of recordings was learned and used for recovering a rain corrupted dataset. Similar to Ruiz-Muñoz et al. (2018), grassmann and tangent singular spectrum analysis have been used for bioacoustic signal classification (Souza et al. 2018, 2019), which can also be extended for spectrogram denoising.



**Fig. 4** Key steps in the AED method (Xie et al. 2015): **a** original spectrogram, **b** spectrogram after Wiener filtering and noise reduction, **c** AED results, **d** AED results after separating large events and removing small events

Most denoising algorithms such as spectral subtraction need to generate a “noise profile”, which are used for enhancing the noisy signal. When dealing with bioacoustic recordings with a short duration, it might be suitable to use a fixed “noise profile”. However, most environmental recordings are continuous with a long duration (longer than 24 hours), and the background noise of those recordings changes over times. Therefore, accurately selecting an approximate “noise profile” is a key to the success of those methods.

Wavelet-based methods have been continuously developed in bioacoustic denoising, and the performance is often better than other methods. However, the efficiency of wavelet-based methods has been argued in a recent study (Brown et al. 2017) when compared to MMSE-STSA. Therefore, improving the efficiency without decreasing the performance need to be further investigated for wavelet-based methods.

After transforming 1D bioacoustic waveform into 2D representations, image analysis techniques which treat the 2D representation as an image provide a range of methods for reducing noise in bioacoustic recordings. Most previous studies use the spectrogram as the 2D representation for the subsequent analysis. However, STFT has a trade-off between time and frequency resolution which is determined by the selected window function, other 2D representation such as wavelet transform, Wigner-ville transform, Hilbert-huang transform can be investigated for bioacoustic signal denoising.

Regarding the computational cost of the aforementioned methods, we can characterize them according to the algorithmic complexity using Big  $O$  asymptotic notation. FIR filtering methods such as low-, band- or high-pass filters have constant computational

complexity  $O(1)$ , meaning that these do not depend on the length of the analyzed signal. The spectral methods based on the Fourier transform, FFT or STFT, have at least  $O(n \log n)$  complexity plus the constants related to iterative filtering procedures, where  $n$  is the size of the input signal. The methods based on vector subspace theory (Colonna and Nakamura 2018) have an initial complexity of  $O(L^3)$  during the adjustment of the filter coefficients, where  $L$  is the number of eigenvalues. After the adjustment was performed, this method has constant complexity  $O(1)$ , equals the FIR filters. Finally, the complexity of the methods based on Deep Neural Networks is approximate  $O(n^4)$  during the forward pass, being  $n$  the number of neurons in the network, where this factor can be in the order of millions depending on the neural network architecture employed. Therefore, we can see that the performance of the methods is strongly related to their computational complexity, i.e., the better the performance, the higher the cost.

Different from denoising bioacoustic recordings for improving the recognition or classification performance, another research direction is to consider the background noise as a source of variation. Therefore, the background noise is regarded as one class for building the classification system, which can also help avoid over-fitting of the classifier. In particular, this kind of method aims to classify noise as one type of class rather than filtering out noise. Luque et al. (2018) proposed a non-sequential approach for classifying anurans. Here, the bioacoustic recordings were segmented into very small pieces of sounds (frames), which was then classified without considering preceding or subsequent information. Xie et al. (2020) proposed to use 1D-CNN for classifying frog species, where background noise was also regarded as one type of class to be classified.

To investigate the described denoising methods, some data sources where bioacoustic data can be obtained including (1) the Historic Marine Mammal Sound Archive<sup>2</sup>, (2) Ecosounds website which is run by the QUT Ecoacoustics Research Group<sup>3</sup>, (3) Anuran calls<sup>4</sup>, (4) MLSP 2013 Bird Classification Challenge<sup>5</sup>, (5) Avian data from New Zealand<sup>6</sup>, and (6) Xeno-canto<sup>7</sup>. However, it is worth noting that the recordings from aforementioned websites are not intuitively obtained for investigating bioacoustic signal denoising algorithms. Therefore, a clear problem of those recordings is the unavailability of truly clean audio recordings to compare the denoising results. One possible solution is to compare the denoised recordings to audio files processed using an aggressive spectral noise gating approach, with noise profiles specifically selected for each recording (Brown et al. 2017).

In summary, for bioacoustic signal analysis, noise reduction plays an important role in the pre-processing part, especially for the machine learning based methods. However, the use of signal denoising might not help the classification when using deep learning based methods (Kong et al. 2017). For the potential researcher who are interested in bioacoustic signal analysis, we have the following suggestions:

- (1) It is worth investigating low-, band- or high-pass filtering method first due to their simplicity and easy implementation, especially dealing with animals species without overlapped frequency range in recordings.

<sup>2</sup> <https://cis.whoi.edu/science/B/whalesounds/index.cfm>.

<sup>3</sup> <https://www.ecosounds.org/>.

<sup>4</sup> <https://archive.ics.uci.edu/ml/datasets/Anuran+Calls+%28MFCCs%29>.

<sup>5</sup> <https://www.kaggle.com/c/mlsp-2013-birds/data>.

<sup>6</sup> <http://www.avianz.net/>.

<sup>7</sup> <https://en.wikipedia.org/wiki/Xeno-canto>.

- (2) For achieving a trade-off between accuracy and efficiency, wavelet-based methods can be a good candidate. Besides wavelet-based methods, the combination and filter-based methods and MMSE-STSA can be another solution.
- (3) When dealing with simultaneously vocalizing animal species, the techniques such as NMF, SS, and AED are worth being investigated.
- (4) If the labels of large volumes of acoustic data are available, the use of deep learning should be considered as an option, especially when the computational cost is less important for building an automatic recognition/classification system.

## 7 Conclusions

In this survey, we mainly aim to provide a research direction for denoising acoustic signals, especially animal sounds. Since signal denoising plays an important role in improving the performance of the whole animal analysis system, various methods have been developed for studying different animal species according to their specific call patterns. All those methods can be grouped into three types: time domain, frequency domain, and time-frequency domain, including (1) Optimal FIR filter, (2) spectral subtraction, (3) MMSE-STSA, (4) wavelet-based denoising, (5) image processing based noise reduction, (6) deep learning based noise reduction. After reviewing the references, we can find that there is no one method that can be successfully used for all kinds of animal calls. For instance, filtering methods such low-, band- or high-pass filtering are efficient, but cannot denoise the bioacoustic recording when the frequency range of target species are overlapped. Wavelet-based denoising methods have high performance in improving the SNR of bioacoustic recordings, but the methods are often less efficient. Furthermore, the metrics consist of subjective and objective evaluations. Choosing the suitable metrics is highly dependent on the particular research problem. Finally, it is necessary to choose an appropriate denoising method based on the interested animal species, background noise, and the real-time requirement.

In general, bioacoustic signal denoising is still in its infancy as a field of study, and potential applications and unsolved problems are extending every day. For future work, it is worth further improving both accuracy and efficiency of bioacoustic signal denoising methods, especially including deep learning techniques. Since many methods have been developed for bioacoustic recordings, another research direction is to develop an algorithm for automatically selecting the most suitable algorithm for the specific recordings. In addition, it is often difficult to obtain the clean signal, which makes it difficult to accurately evaluate signal denoising algorithms.

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